

Tamminana Sumith Kumar

AI Engineer | GenAI & Agentic Systems | Software Developer Engineer

linkedin.com/in/tamminana-sumith-kumar | github.com/SumithKumar765 | Portfolio | +91-7659833962 | kumarsumith98765@gmail.com

EDUCATION

B.Tech (Hons) in ECE — Centurion University

2020 – 2024 | GPA: 8.64/10

Dean's Best Student Award | Coursework: DSA, Machine Learning, Digital Signal Processing

PROFESSIONAL EXPERIENCE

Webassic — Software Developer Engineer

Mar 2025 – Present

Boxtron ERP

Apr – Sep 2025

Frappe Framework · Python · JavaScript · ERPNext · MariaDB · Redis Cache · Redis Queue · WebSockets · REST API · India Compliance · GST

- Built custom ERPNext app on Frappe (Python) with 35+ RBAC roles for end-to-end business workflow automation and GST compliance.
- Engineered GST-compliant financial modules: invoice generation, delivery/returnable/return-back challans via India Compliance integration.
- Optimized with Redis Cache + Queue for async job processing; WebSockets for real-time notifications; custom REST APIs in Python & JS.

Apartment Explorers

Sep – Oct 2025

Django · Python · PostgreSQL · HTML5 · Bootstrap CSS · REST API · RBAC · OWASP · Agile

- Delivered full-stack Django + PostgreSQL property platform (MagicBricks-style) from scratch to production in 1 month using Agile.
- Architected 3-tier RBAC: Admin approval dashboard · Property-owner listing portal (condo/apartment/townhouse/single-house) · End-user search.
- Achieved 80% security compliance per client R&D audit — OWASP-aligned auth, input validation, and secure session management.

TECHNICAL PROJECTS

Autonomous Healthcare Voice AI Receptionist | Real-Time Conversational AI System

june-2026

Python · LiveKit WebRTC · Deepgram (STT) · Groq (Llama 3.3) · Cartesia (TTS) · Simli · FastAPI · Supabase · Async Python · Real-Time Voice Pipeline

- Architected an end-to-end autonomous healthcare receptionist from PRD to production, integrating LiveKit WebRTC, Deepgram, Groq (Llama 3.3), Cartesia, and Simli to achieve a state-of-the-art sub-600ms end-to-end response latency (Time-to-First-Audio) for seamless, human-like conversational pacing.
- Built a robust FastAPI and Supabase backend that handles secure signaling and state synchronization while custom-tuning asynchronous Python loops to hit strict latency budgets: ~150ms for STT, <200ms for LLM Time-to-First-Token, and ~100ms for TTS rendering, successfully deploying the architecture on free-tier cloud environments.

Propsense | AI-Powered Property Renovation Platform

may-2026

React 19 · Vite · FastAPI · Google Gemini API · Docker · Nginx · Cloudflare Pages · Render · jsPDF · Prompt Engineering

- Built Propsense, a full-stack AI-powered property renovation platform using React 19, Vite, FastAPI, and Google Gemini API that automates architectural region detection from property images, maps materials via an interactive UI, and generates real-time cost estimates — deployed on Cloudflare Pages + Render (Docker + Nginx) with a PDF export feature using jsPDF.
- Engineered an end-to-end REST API backend with Python/FastAPI, implemented AI prompt engineering for structured Gemini responses, researched and evaluated 3 image generation models (instruct-pix2pix, Stable Diffusion, Fal.ai) for AI visualization, and delivered a production-ready prototype with load balancing and modular architecture ready for JWT auth and GPU-based redesign integration.

AI Elite Travel Agent

Mar-2026

FastAPI · LangGraph · Groq · Llama-3.3-70b · LLM · SQLite · OAuth 2.0 · SMTP · Leaflet.js · SQLAlchemy · SSE

- Engineered agentic travel backend with LangGraph multi-node LLM workflow; cut latency from 15s+ to <3s via Groq inference.
- Optimized backend data pipelines by integrating FastAPI and SSE (Server-Sent Events) streaming, successfully slashing LLM response latency by 94% (from 50 seconds down to under 3 seconds).
- Integrated OAuth 2.0 + SMTP OTP auth, Leaflet.js mapping, Open-Meteo weather, PDF export & Google Calendar (.ics) sync.

TECHNICAL SKILLS

Languages: Python (Expert) · JavaScript · SQL · C++ · HTML5/CSS3, Pytorch.

AI / ML: GenAI · Agentic AI · ML · LangGraph · LangChain · Groq · LLMs (Llama 3.3, Gemini, open ai, Ollama, Gemma3, Nano Banana) · Vector DBs(pinecone, Fiass) · LoRA · Fine-tuning · Prompt Engineering · NLP · Real-Time Voice AI (VAD, STT/TTS pipelines)

Web & Backend: FastAPI · Django · Frappe Framework · ERPNext · REST API · WebSockets · SSE · Redis Cache · Redis Queue · Bootstrap CSS · React

DevOps & Cloud: Docker · Nginx · Cloudflare Pages · Render · Vercel · Git/GitHub · OAuth 2.0 · SMTP · Linux

Databases: PostgreSQL · MariaDB · MYSQL · Supabase

Tools: Power BI · Excel · Postman · Leaflet.js

CERTIFICATIONS & ACHIEVEMENTS

Generative AI Specialization — Coursera | AWS IoT Analytics & Greengrass — Edge Computing & Data Processing | Dean's Best Student Award